

2018EBN122C2**CLASS TEST**

Kopiereg voorbehou

21 September 2018

2018EBN122C2**KLASTOETS**

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21 September 2018

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form <i>30 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		8 + OCR	

Important

- Die eksamenregulasies van die Universiteit van Pretoria geld
The examination regulations of the University of Pretoria apply.
- **All questions must be answered on the Optical Character Recognition (OCR) form.**
Alle vrae moet op die Optiese Karakter Herkenning ('OCR')- vorm ingevul word.
- Read the instructions on the OCR form carefully.
Neem noukeurig kennis van die instruksies op die 'OCR'-vorm
- Answers must include units!
Antwoorde moet eenhede insluit!
- **Final answers must be written as real numbers and not as fractions, and must be rounded to 5 significant numbers**
Finale antwoorde moet as reële getalle geskryf word en nie as breuke nie, en moet afgerond word na 5 beduidende syfers.

Doseent(e) / Lecturer(s): D le Roux, JP de Villiers, JP Jacobs

INSTRUCTIONS**Do step 0 in the first 5 minutes of the session. (0-5 min)****Do step 1 in the first 30 minutes of the test. (5-35 min)****Do step 2 within the following 10 minutes. (35-45 min)****STEP 0 (-5 min to 0 min):** Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is:

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- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	-	1	.	4	5					μ	A						
02	3	4	0							V							
03	1	5	.	2						k							
04	2	0								Ω							

INSTRUKSIES

Doen stap 0 in die eerste 5 minute van die *sessie*. (0-5 min)

Doen stap 1 in die eerste 30 minute van die *toets*. (5-35 min)

Doen stap 2 binne die daaropvolgende 10 minute. (35-45 min)

STAP 0 (-5 min to 0 min): Voltooi die voorblad van die *OCR*-vorm

STAP 1 (0 min to 30 min): Doen jou eie berekeninge op die *VRAESTEL*. Die *VRAESTEL* word **nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die *OEFEN-ANTWOORDBLAD*. Maak seker om eenhede in te sluit. Die *OEFEN-ANTWOORDBLAD* word **nie** ingeneem nie.

STAP 2 (35 min to 40 min): Dra die antwoorde vanaf die *OEFEN-ANTWOORDBLAD* oor na die *OCR*-vorm.

- Die *Assessment ID* is:

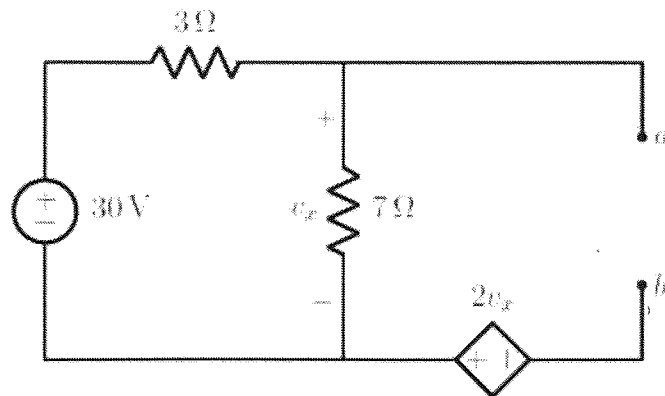
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- Gebruik 'n meganiese potlood met 0.5mm HB lood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak asb. gebruik van punte.

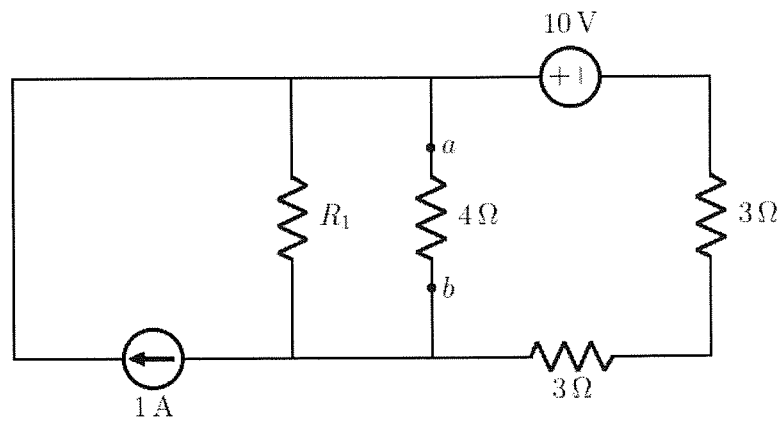
Voorbeelde

01	-	1	.	4	5					m	A							
02	3	4	0							V								
03	1	5	.	2														
04	-	1								Ω								

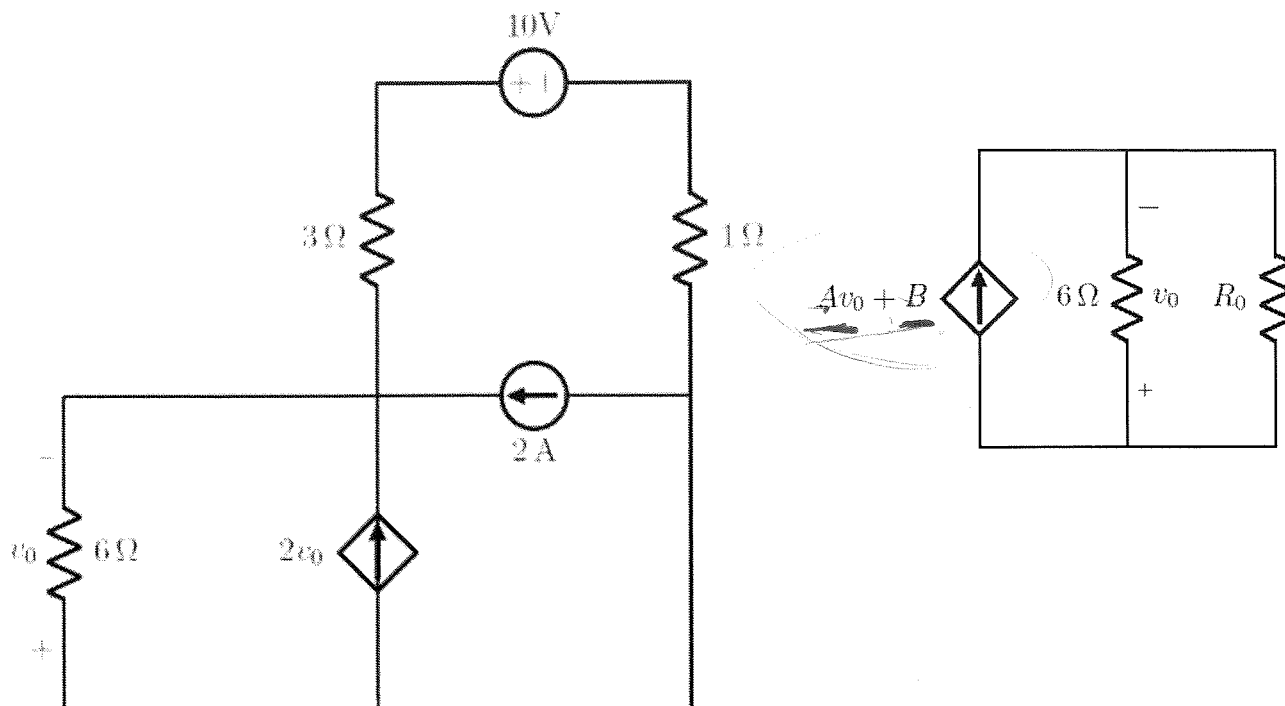
Question 1-2 [4]		Vraag 1-2 [4]	
Determine the Thevenin equivalent voltage V_{Th} and resistance R_{Th} with reference to terminals a - b .		Bepaal die Thevenin ekwivalente spanning V_{Th} en weerstand R_{Th} met verwysing tot terminale a en b .	
01	V_{Th}	01	V_{Th}
02	R_{Th}	02	R_{Th}



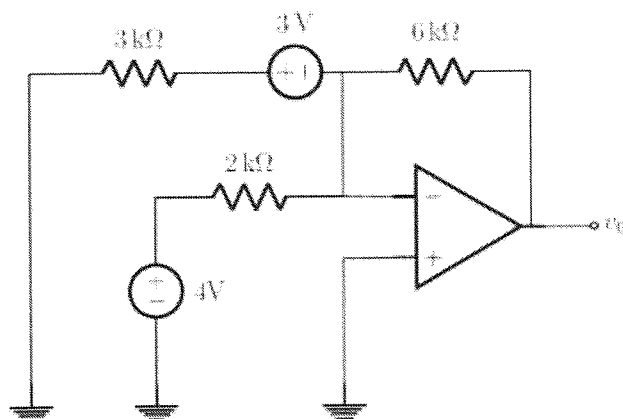
Question 3 [2]		Vraag 3 [2]	
03	It is known that maximum power is delivered to the $4\ \Omega$ load resistor. Determine R_1 .	03	Dit is bekend dat maksimum drywing oorgedra word aan die $4\ \Omega$ lasweerstand. Bepaal R_1 .



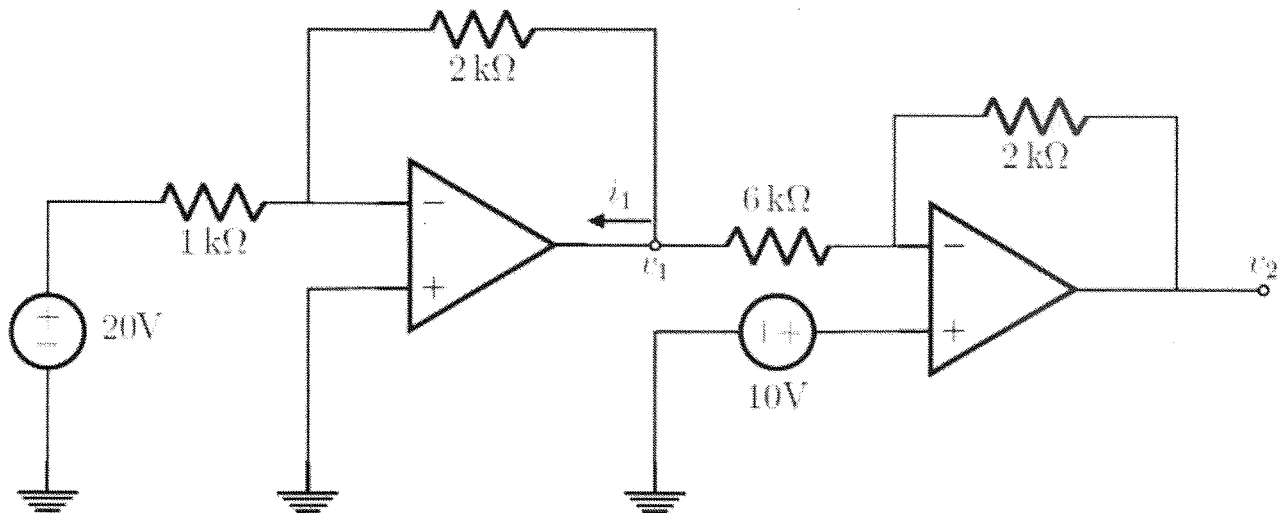
Question 4 - 6 [4]		Vraag 4 - 6 [4]	
Simplify the circuit on the left by means of source transformation so that it looks like the one on the right. Determine:		Vereenvoudig die stroombaan aan die linkerkant met behulp van brontransformasie sodat dit lyk soos die kringbaan aan die regterkant. Bepaal:	
04	R_o [1]	04	R_o [1]
05	A [1]	05	A [1]
06	B [2]	06	B [2]



Question 7 [2]		Vraag 7 [2]	
07	Determine v_o . Assume that the op-amp is ideal.	07	Bepaal v_o . Aanvaar dat die OV ideaal is.



Questions 8-9 [4]		Vraag 8-9 [4]	
Two ideal op-amps are interconnected as indicated. It is known that $v_I = -40$ V.		Twee ideale OV's word verbind soos aangetoon. Dit is bekend dat $v_I = -40$ V.	
08	Determine v_2 . [2]	08	Bepaal v_2 . [2]
09	Determine i_I . [2]	09	Bepaal i_I . [2]



**PRACTICE ANSWER SHEET /
OEFEN-ANTWOORDBLAD**

This page may NOT be removed from the test. It will not be taken in.

Assessment ID	2	0	1	8	E	B	N	1	2	2	C	2		
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Answer / Antwoord

Unit / Eenheid

Answers must be rounded to 5 significant numbers
Antwoorde moet afgerond word na 5 beduidende syfers.

01	$V_{Th} =$		
02	$R_{Th} =$		
03	$R_I =$		
04	$R_0 =$		
05	$A =$		
06	$B =$		
07	$v_0 =$		
08	$v_2 =$		
09	$i_I =$		